

WIPRO PYTHON AUTOMATION COURSE

MODULE_2_CODE_REPORT

UNIT TEST FRAMEWORK

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Course:- Python Automation Testing

Technology:- Unittest, PyTest Framework, Page Object Model

Enrolment No. :- 12023052018002

Stream:- CSBS

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FOLDER STRUCTURE IN MY VSCODE

Selenium_Module_2 – Project Folder Structure

(Page Object Model | Data-Driven Testing | PyTest Framework)

Folder Structure	Folder / File	Type	Purpose / Description
 Selenium_Module_2	Selenium_Module_2	Root Folder	Main project folder containing all files and subfolders.
 .venv	.venv	Folder	Python virtual environment (auto-generated). Stores installed packages.
 data	data	Folder	Contains external test data for data-driven testing.
 login_data.csv	login_data.csv	File (CSV)	Contains test cases (username, password, expected results, error messages).
 pages	pages	Folder	Contains Page Object classes with locators and UI methods.
 login_page.py	login_page.py	File (Python)	Login page locators and methods.
 dashboard_page.py	dashboard_page.py	File (Python)	Dashboard page locators and methods.
 tests	tests	Folder	Contains actual PyTest test cases.
 test_login.py	test_login.py	File (Python)	Data-driven login test cases with assertions.
 utils	utils	Folder	Contains utility/helper functions.
 test_data.py	test_data.py	File (Python)	Reads and processes the CSV test data using pandas.
 reports	reports	Folder	Stores the test execution reports.
 report.html	report.html	File (HTML)	Generated HTML report (after test execution).
 screenshots	screenshots	Folder	Stores screenshots of failed test cases.
 (failed test screenshots)	(failed test screenshots)	Image Files	Screenshots captured automatically on test failure (by conftest.py).
 config.py	config.py	File (Python)	Contains application URL, browser type, and wait configuration.
 conftest.py	conftest.py	File (Python)	PyTest fixtures for browser setup, teardown, and failure screenshots.
 pytest.ini	pytest.ini	File (INI)	PyTest configuration (test paths, markers, options).
 requirements.txt	requirements.txt	File (Text)	List of required Python packages.
 .gitignore (optional)	.gitignore (optional)	File (Text)	Specifies files/folders to ignore in Git (optional).

ASSIGNMENT 7 — PAGE OBJECT MODEL

Requirements: -

Take working scripts and restructure them into Page Object Model. Create separate page classes containing locators and UI methods. Keep assertions separate from locators.

Folder Name: -

pages/login_page.py

Code: -

```
from selenium import webdriver

from pages.login_page import LoginPage

from pages.dashboard_page import DashboardPage

def test_valid_login():

    # Start browser

    driver = webdriver.Chrome()

    # Open SauceDemo

    driver.get("https://www.saucedemo.com/")

    # Create LoginPage object

    login_page = LoginPage(driver)

    # Perform login

    login_page.login(

        "standard_user",

        "secret_sauce"

    )

    # Create DashboardPage object

    dashboard_page = DashboardPage(driver)

    # Assertions

    assert "/inventory.html" in driver.current_url
```

```

assert dashboard_page.is_inventory_displayed()

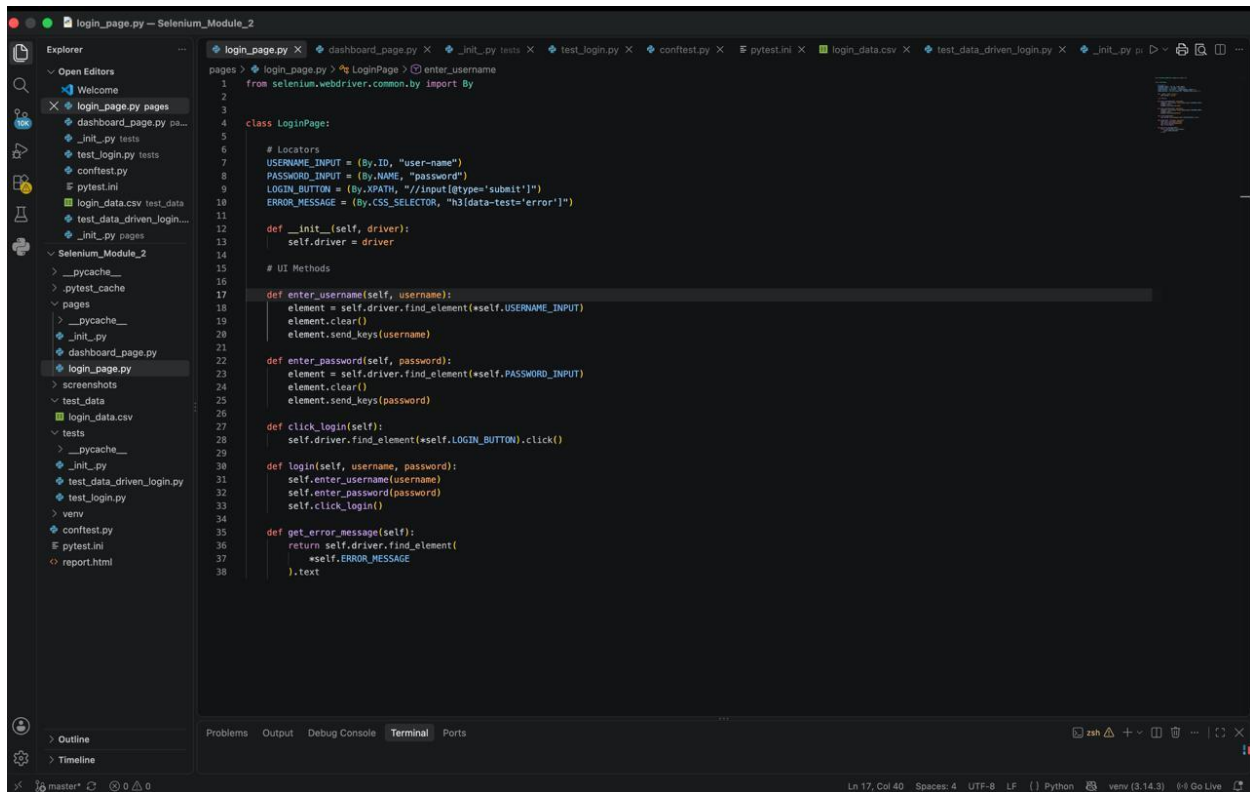
assert dashboard_page.get_page_title() == "Products"

# Close browser

driver.quit()

```

Screenshots: -



Folder Name: -

pages/dashboard_page.py

Code: -

```

from selenium.webdriver.common.by import By

from selenium.webdriver.support.ui import WebDriverWait

from selenium.webdriver.support import expected_conditions as EC

class DashboardPage:

    # Locators

    TITLE = (By.CLASS_NAME, "title")

```

```
INVENTORY_CONTAINER = (By.ID, "inventory_container")
```

```
def __init__(self, driver):
```

```
    self.driver = driver
```

```
    self.wait = WebDriverWait(driver, 10)
```

```
# UI Methods
```

```
def get_page_title(self):
```

```
    return self.wait.until(
```

```
        EC.visibility_of_element_located(self.TITLE)
```

```
    ).text
```

```
def is_inventory_displayed(self):
```

```
    return self.wait.until(
```

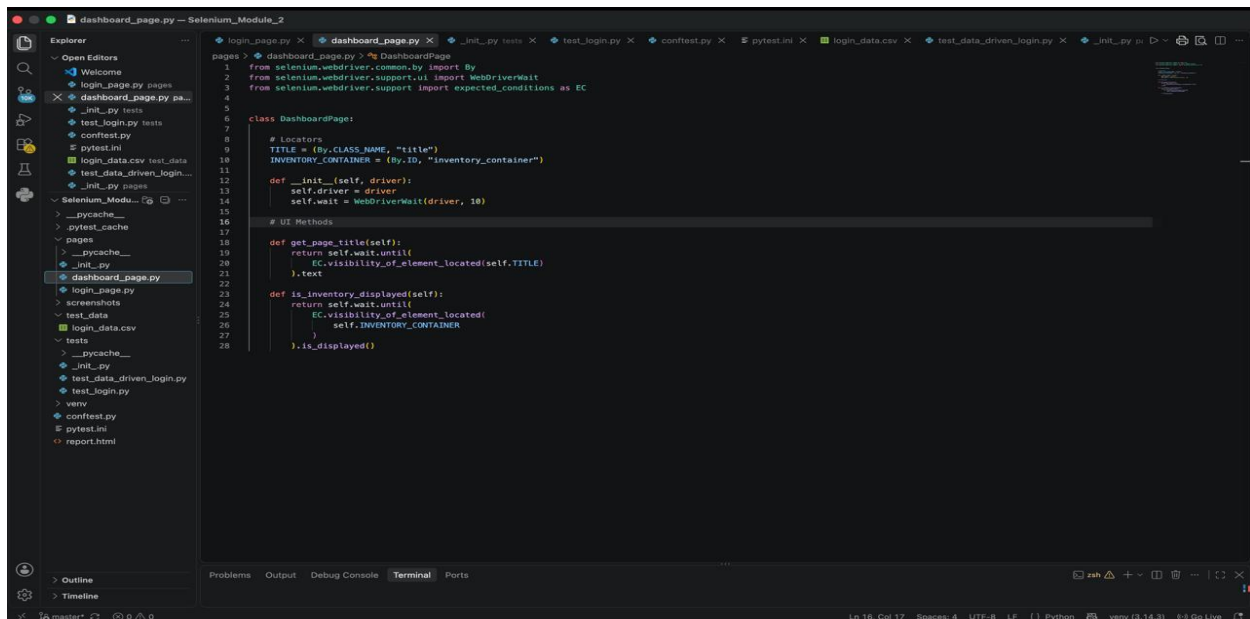
```
        EC.visibility_of_element_located(
```

```
            self.INVENTORY_CONTAINER
```

```
        )
```

```
    ).is_displayed()
```

Screenshots: -



ASSIGNMENT 8 — DATA-DRIVEN TESTING

Requirements: -

Build a login script that reads multiple test cases from an external source such as Excel, JSON or CSV.

Folder Name: -

data/login_data.csv

Code: -

test_case,username,password,expected_result,expected_error

TC001,standard_user,secret_sauce,success,

TC002,standard_user,wrong_password,error,Epic sadface: Username and password do not match any user in this service

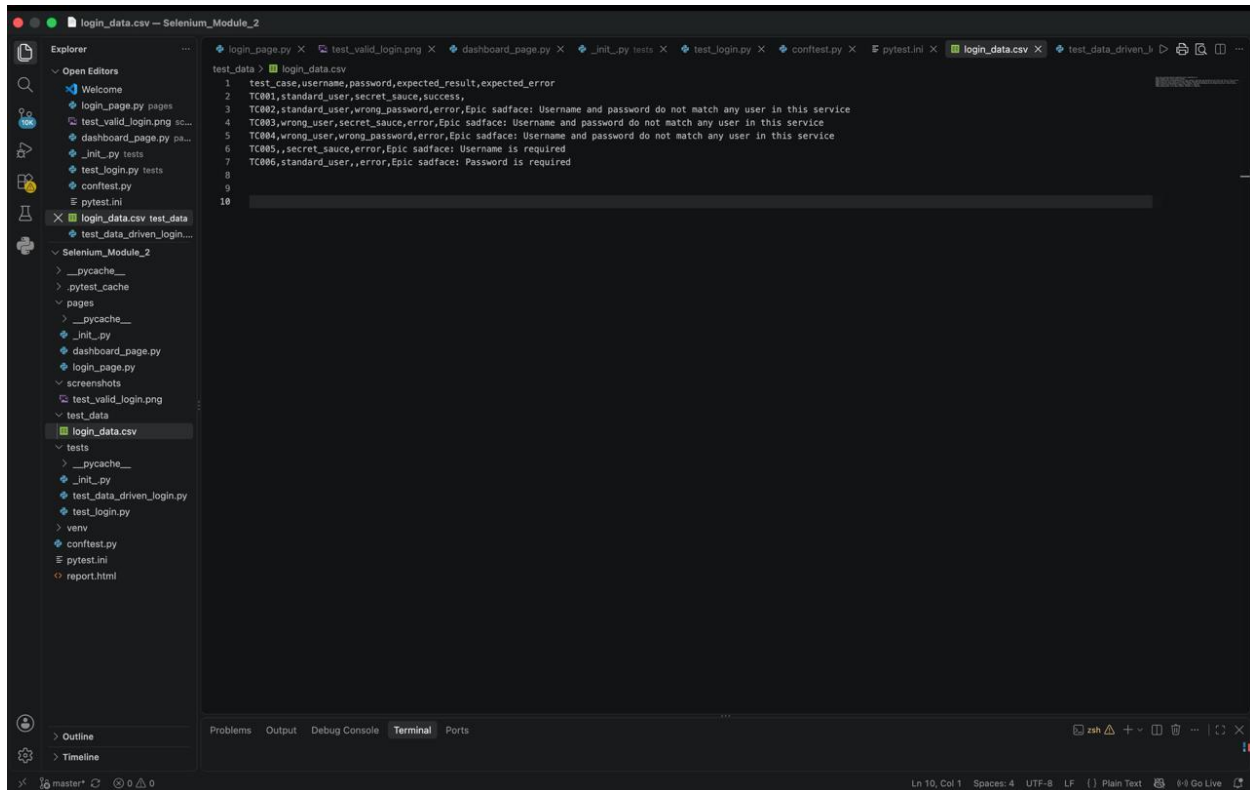
TC003,wrong_user,secret_sauce,error,Epic sadface: Username and password do not match any user in this service

TC004,wrong_user,wrong_password,error,Epic sadface: Username and password do not match any user in this service

TC005,,secret_sauce,error,Epic sadface: Username is required

TC006,standard_user,,error,Epic sadface: Password is required

Screenshots: -



Folder Name: -

utils/test_data.py

Code: -

```
from pathlib import Path

import pandas as pd

DATA_FILE = (
    Path(__file__).resolve().parent.parent
    / "data"
    / "login_data.csv"
)

def load_login_data():
    df = pd.read_csv(
        DATA_FILE,
```

```

        dtype=str,
        keep_default_na=False
    )
    required_columns = {
        "test_case",
        "username",
        "password",
        "expected_result",
        "expected_error"
    }
    missing = required_columns - set(df.columns)
    if missing:
        raise ValueError(
            f"Missing columns: {sorted(missing)}"
        )
    return df.to_dict(orient="records")

```

Folder Name: -

tests/test_login.py

Code: -

```

from pages.login_page import LoginPage
from pages.dashboard_page import DashboardPage

def test_valid_login(driver):
    # Open SauceDemo
    driver.get("https://www.saucedemo.com/")

    # Create LoginPage object
    login_page = LoginPage(driver)

```

```
# Perform login
```

```
login_page.login(  
    "standard_user",  
    "secret_sauce"  
)
```

```
# Create DashboardPage object
```

```
dashboard_page = DashboardPage(driver)
```

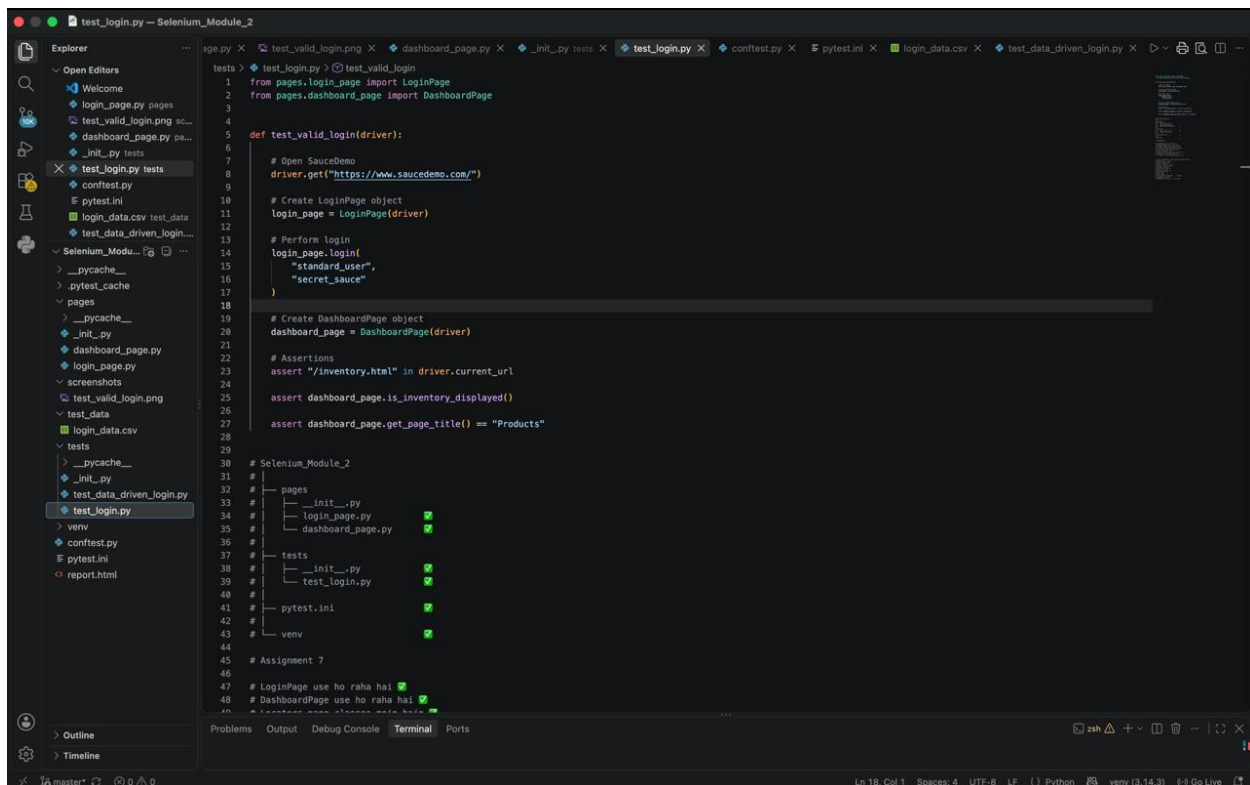
```
# Assertions
```

```
assert "/inventory.html" in driver.current_url
```

```
assert dashboard_page.is_inventory_displayed()
```

```
assert dashboard_page.get_page_title() == "Products"
```

Screenshots: -



ASSIGNMENT 9 – PYTEST + HTML REPORTING

Requirements: -

Convert setup to PyTest, use PyTest fixtures for driver initialization and teardown, and auto-generate HTML execution report with screenshots of failed test steps.

File Name: -

conftest.py

Code: -

```
import os

import pytest

import pytest_html

from selenium import webdriver

@pytest.fixture
def driver():

    # Start Chrome

    driver = webdriver.Chrome()

    # Open browser maximized

    driver.maximize_window()

    yield driver

    # Close browser

    driver.quit()

@pytest.hookimpl(hookwrapper=True)
def pytest_runtest_makereport(item, call):

    # Run the actual test

    outcome = yield
```

```

# Get test report
report = outcome.get_result()

# Only capture screenshot when the actual test fails
if report.when == "call" and report.failed:

    # Get driver from fixture
    driver = item.funcargs.get("driver")

    if driver:

        # Create screenshots folder
        os.makedirs("screenshots", exist_ok=True)

        # Screenshot filename
        screenshot_name = (
            f"{item.name}.png"
        )

        screenshot_path = os.path.join(
            "screenshots",
            screenshot_name
        )

        # Save screenshot to file
        driver.save_screenshot(screenshot_path)

        # Read screenshot as binary data
        with open(screenshot_path, "rb") as image_file:
            screenshot_data = image_file.read()

        # Get existing HTML report extras
        extras = getattr(report, "extras", [])

        # Attach screenshot to HTML report
        extras.append(

```

```

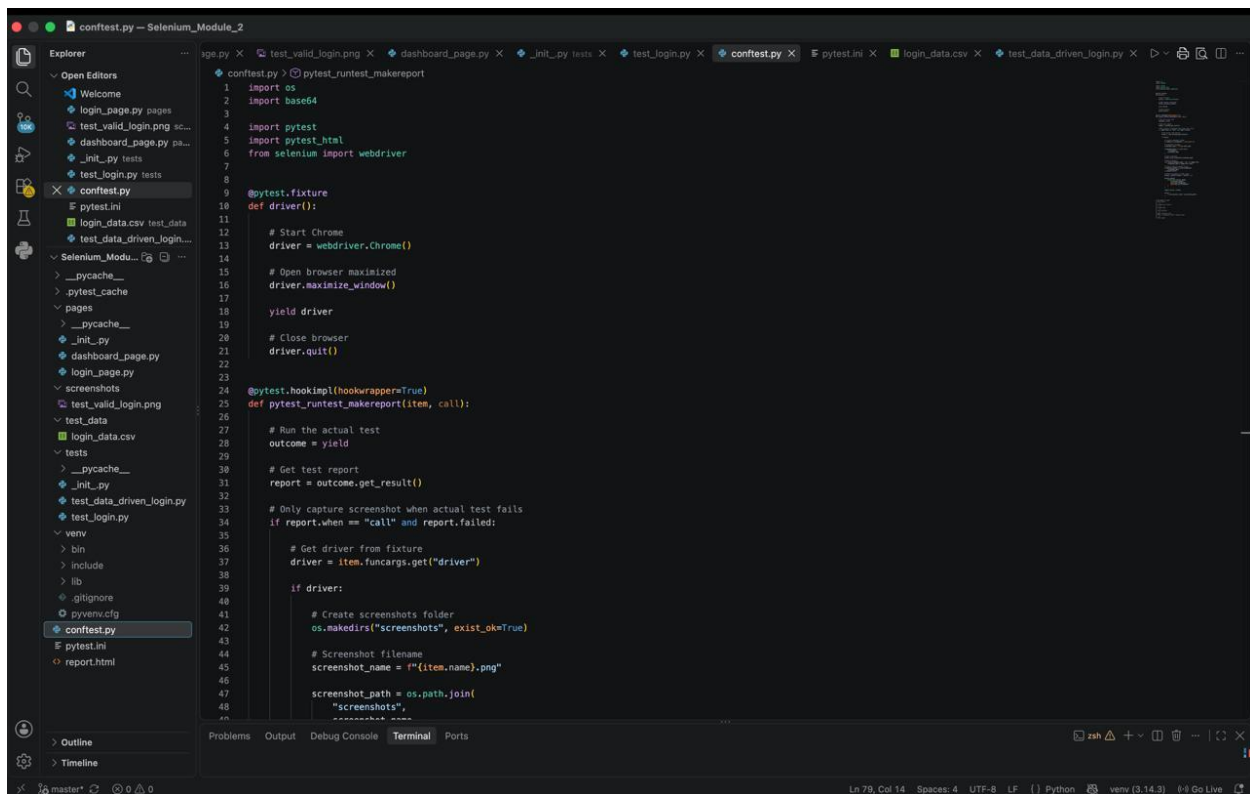
    pytest_html.extras.image(
        screenshot_data,
        mime_type="image/png",
        name="Failure Screenshot"
    )
)

# Store extras back into report
report.extras = extras

print(
    f"\nScreenshot saved: {screenshot_path}"
)

```

Screenshots: -



File Name: -

requirements.txt

Code: -

selenium

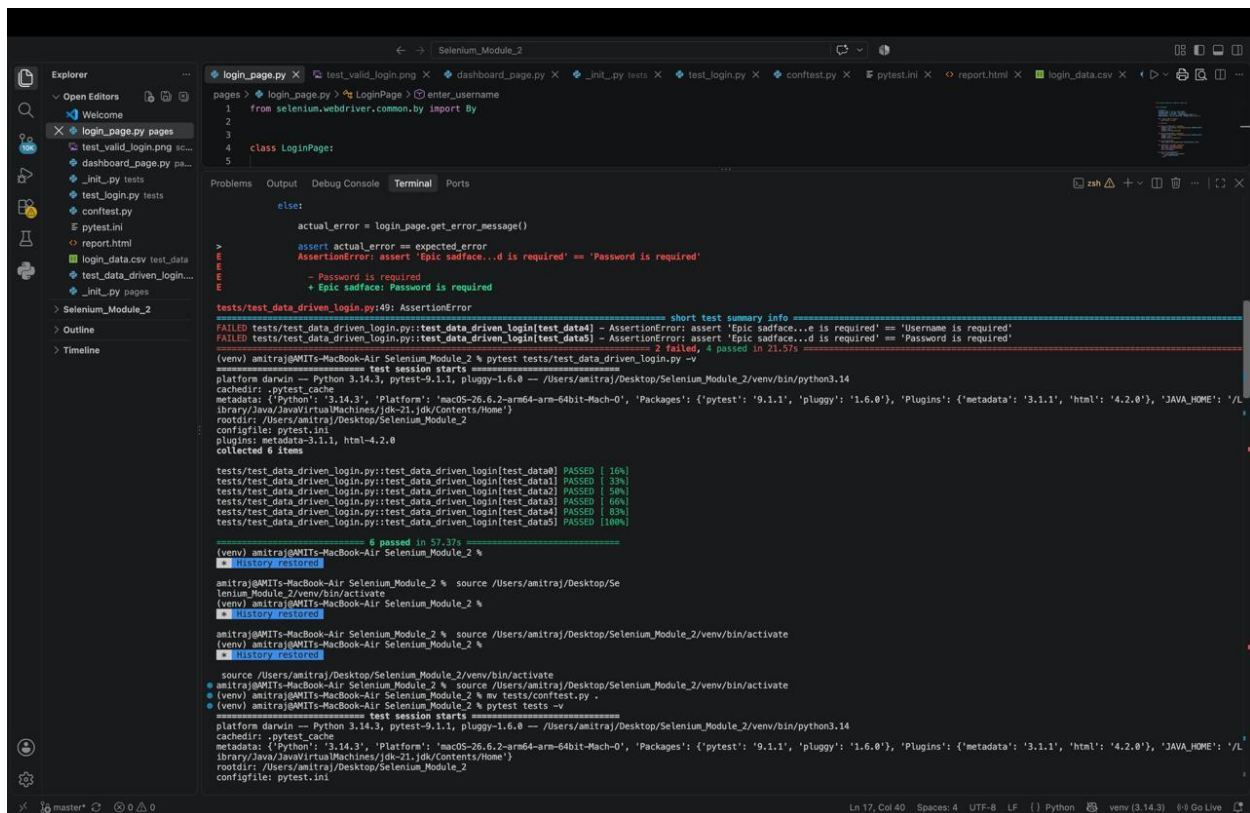
pytest

pytest-html

pandas

openpyxl

Code Execution Proof: -



The screenshot shows a VS Code editor with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named 'Selenium_Module_2' with files like 'login_page.py', 'test_valid_login.png', 'dashboard_page.py', and 'test_login.py'. The terminal displays the output of running 'pytest tests/test_data_driven_login.py -v'. The output shows that the test session starts, and the test 'test_data_driven_login' is executed. The test passes, and the output shows '6 passed in 57.37s'. The terminal also shows the command 'source /Users/Amiraj/Desktop/Selenium_Module_2/venv/bin/activate' and the command 'mv tests/conf/test.py'.

```
login_page.py X test_valid_login.png X dashboard_page.py X _init_py tests X test_login.py X conf/test.py X pytest.ini X report.html X login_data.csv X
pages > login_page.py > LoginPage > enter_username
1 from selenium.webdriver.common.by import By
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4 class LoginPage:
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